MTH 211: Theory of Statistics

Lecture 36: Bayes' Risk, Wtd. Squared Error Loss fn

Recall: If δ_B is a Bayes estimator (wrt a loss fn and a prior) then the average risk $r(\pi,\delta)$ is minimized at δ_B and this minimum risk is termed as the Bayes' risk.

Ex: Calculation of Bayes risk from the following set-up:

$X \sim Bin(n,\theta)$, $\theta \in [0,1]$, $\mathcal{X} = \{0,1,\dots,n\}$

$L(\theta,d) = (\theta-d)^2$ (squared error loss fn)

$\theta \sim U(0,1)$ (non-informative prior $U(0,1)$)

We've already obtained $\delta_{Bayes}(X) = \dfrac{X+1}{n+2}$.

To obtain the Bayes' risk

$$r(\pi,\delta_{Bayes}) = \int_{\Theta}\int_{\mathcal{X}} L(\theta,\delta_{Bayes}) f(x|\theta)\,dx\,\pi(\theta)\,d\theta$$

$$= \int_0^1 \sum_{x=0}^{n} \left(\theta - \frac{x+1}{n+2}\right)^2 f(x|\theta)\,d\theta$$

$$= \int_0^1 E_{X|\theta}\left(\frac{X+1}{n+2} - \theta\right)^2 d\theta = \frac{1}{(n+2)^2}\int_0^1 E_{X|\theta}\left((X^2+2X+1) - ((n+2)\theta)\right)^2 d\theta$$

Use, $E(X) = n\theta$
$E(X^2) = n(\theta(1-\theta)) + n^2\theta^2$
(obtd from the conditional distn $f(x|\theta) \equiv Bin(n,\theta)$)

$$= \frac{1}{(n+2)^2}\int_0^1 \left[n\theta(1-\theta) + (2\theta-1)^2\right] d\theta = \frac{1}{6(n+2)}$$

(Calculated directly from the conditional distn w/o the posterior distn)

Claim: Bayes estimator is a fn of sufficient statistic

$X_1,\dots,X_n$ iid from $f(x|\theta)$, $T(\underline{X})$ suff. for θ

By NFFT, $f(\underline{x}|\theta) = h(\underline{x})\, g_\theta(T(\underline{x}))$, where $h(\underline{x})$ does not depend on θ, fn of $x_1,\dots,x_n$, $g_\theta(T(\underline{x}))$ depends on θ, and on $x_1,\dots,x_n$, but only thro' $T(x_1,\dots,x_n)$, the suff. stat.

Then the posterior distn

$$\pi(\theta|\underline{x}) = \frac{h(\underline{x})\, g_\theta(T(\underline{x}))\,\pi(\theta)}{\int h(\underline{x})\, g_\theta(T(\underline{x}))\,\pi(\theta)\,d\theta}$$

$$= \frac{g_\theta(T(\underline{x}))\,\pi(\theta)}{\int g_\theta(T(\underline{x}))\,\pi(\theta)\,d\theta}, \text{ depends on } x_1,\dots,x_n \text{ only thro' } T(\underline{x})$$

For squared error loss, the mean of the posterior is the Bayes estimator, depends on $x_1,\dots,x_n$ only thro' $T(x_1,\dots,x_n)$.

This is true for other loss fns as well, where Bayes estimator is median or mode from the posterior distn.

A different loss function — the weighted squared error loss function

$L(\theta,d) = w(\theta)(\theta-d)^2, \quad w(\theta)>0$

The posterior risk $E\left[L(\theta,\delta(x))\,|\,X=x\right] = E\left[w(\theta)(\theta-\delta(x))^2\,|\,X=x\right]$, is to be minimized to get the Bayes estimator, δ_B, say.

$$\left.\frac{\partial E_{\theta|x}\left[w(\theta)(\theta-\delta(x))^2\,|\,X=x\right]}{\partial\delta}\right|_{\delta_B} = 0$$

$$\Rightarrow -2E\left[w(\theta)\left(\theta-\delta_B^*(x)\right)\,|\,X=x\right]=0$$

$$\Rightarrow \delta_B^*(X) = \frac{E_{\theta|x}(w(\theta)\theta/X=x)}{E_{\theta|x}(w(\theta)/X=x)}$$

Example

$X\sim P(\lambda), \quad L(\lambda,d)=\dfrac{(\lambda-d)^2}{\lambda}, \quad \Theta=\mathbb{R}^+, \quad \mathcal{X}=\{0,1,\cdots\}.$

$\lambda\sim G(a,b) \quad a,b>0$

$$\Rightarrow \delta_B = \frac{E\left[\frac{1}{\lambda}\cdot\lambda\,|\,X=x\right]}{E\left(\frac{1}{\lambda}\,|\,X=x\right)} = \frac{1}{E\left(\frac{1}{\lambda}\,|\,X=x\right)}$$

Need to find the posterior distn $\pi(\lambda|x)$ first.

$$f(x|\lambda)=\frac{e^{-\lambda}\lambda^x}{x!}, \quad \pi(\lambda)=\frac{1}{\Gamma a\, b^a}e^{-\lambda/b}\lambda^{a-1}$$

so that, $$\pi(\lambda|x)=\frac{f(x|\lambda)\pi(\lambda)}{\int_0^\infty f(x|\lambda)\pi(\lambda)\,d\lambda} \equiv G\left(a+x,\frac{b}{b+1}\right)$$

and $$E_{\lambda|x}\left(\frac{1}{\lambda}\right)=\frac{1}{\Gamma(a+x)\left(\frac{b}{b+1}\right)^{a+x}}\int e^{-\lambda\left(\frac{b+1}{b}\right)}\lambda^{a+x-2}\,d\lambda=\frac{b+1}{b(a+x-1)}$$

$$\Rightarrow \delta_{\text{Bayes}}^{(\text{weighted})}=\frac{b(a+X-1)}{b+1}$$

Whereas, with the squared loss $L(\theta,d)=(\theta-d)^2$,

$$\delta_{\text{Bayes}}=E[\lambda\,|\,X=x]=\frac{b(a+X)}{b+1}.$$

Next, other loss functions, like absolute error, 0-1 loss fns.